Is biodiversity loss in beef-exporting countries associated with the volume of beef exports to the United Kingdom?

1. Introduction

Global agricultural expansion is recognised as one of the leading drivers of biodiversity loss (the reduction in the variety and variability of living organisms within ecosystems), particularly (but not exclusively) through land-use change associated with livestock production. Agriculture occupies approximately 43% of the Earth's ice-free and non-desert land surface and is responsible for around 31% of global greenhouse gas emissions, making it the dominant driver of terrestrial environmental change (Poore and Nemecek 2018). Within this system, livestock production - particularly beef. Despite providing only 17% of global calories and 38% of protein (Ritchie & Roser 2024), animal agriculture uses 83% of global farmland, primarily due to the large land requirements for grazing and feed production (Poore and Nemecek, 2018). This land-use intensity drives biodiversity loss (the decline in species abundance and ecosystem integrity) through deforestation, habitat fragmentation, and land degradation.

Beef production is especially impactful as cattle require significantly more land than other food sources, both directly for grazing and indirectly for crops used as feed. As a result, expanding livestock systems has been suggested to reduce species abundance and the disruption of locally and globally disrupt ecosystems, particularly in biodiverse regions undergoing agricultural expansion like the Amazon (Garcia et al. 2017). At the same time, increasing globalisation has led to the development of complex and interconnected international supply chains, where environmental and other impacts are often geographically separated from consumption. This phenomenon is commonly described as the "externalisation" of environmental costs (the shifting of ecological impacts from consuming to producing countries).

The United Kingdom is a significant importer of agricultural products, including beef (GOV.UK 2025). Raising an important question regarding whether patterns of consumption in the UK are associated with biodiversity outcomes in exporting nations. To answer and address this there must be linking of ecological indicators with international trade flows, which can prove to be challenging due to problems in data collection.

Regardless, this project aims to investigate the research question: "Is biodiversity loss in beef-exporting countries associated with the volume of beef exports to the United Kingdom?" To

examine this, there was combination of country-level biodiversity data, measured using the Biodiversity Intactness Index (BII: The average abundance of native species in a given area, compared to their abundances in an intact, undisturbed ecosystem), with bilateral trade data on beef exports to the UK. By constructing both panel datasets (observations across countries and time) and cross-sectional averages (country-level summaries), the study evaluates whether higher export volumes are systematically (widespread rather than isolated) associated with greater biodiversity loss. However, such relationships are unlikely to be straightforward, as biodiversity loss is influenced by multiple interacting factors, including domestic agricultural practices, all levels of policy, and broader global market dynamics.

2. Data and Methodology

2.1 Data Sources

Data sources comprised of integrating two primary datasets to analyse the relationship between international trade and environmental health. The first is FAOSTAT trade data, which records the specific tonnage of beef the UK imports from global suppliers. The second is the BIO dataset, which provides the Biodiversity Intactness Index (BII), a measure of how much of a country's original nature remains. By combining them into a single file, we can directly compare the amount of beef the UK buys from a country against that same country's biodiversity loss score. This allows us to move from general trade figures to a specific assessment of where the UK's consumption can be overlapped with ecological degradation. Ecological health is measured using the Biodiversity Intactness Index (BII), sourced from the BIO dataset. The analysis utilises historical data (2010 to 2014) and projected data (2015 to 2020) under the Shared Socioeconomic Pathway 2 (SSP2) [a "middle of the road" scenario where social, economic, and technological trends do not shift markedly from historical patterns]. These data are paired with bilateral trade statistics from FAOSTAT [the Food and Agriculture Organization Corporate Statistical Database], specifically annual "Beef and Veal" export quantities, measured in tonnes, from global reporting countries to the United Kingdom.

Data range: the analysis uses data spanning 2001-2024, including historical records and projections (SSP2 scenario) to maximize sample coverage and capture long-term trends.

2.2 Data Preparation and Decision-Making

Preparation steps to make the datasets work well combined include addressing naming inconsistency across international databases, for instance the distinction between “China, mainland” and “China”. To ensure accuracy and prevent the exclusion of major exporters, a custom mapping dictionary was developed to harmonise all reporting entities to ISO3 alpha-3 codes [standardized three letter country codes like BRA for Brazil or ARG for Argentina]. This standardisation was needed to ensure that the ecological data for a specific territory mapped to its corresponding trade data.

Furthermore, a specific variable for biodiversity loss (bii_loss) was made, calculated as 1 minus BII. While the raw index measures intactness where 1.0 is perfect. A positive relationship in the regression now directly represents an association between trade and degradation rather than intactness. As the original BII is a scale where 1.0 is perfectly intact nature. Using it raw would produce a negative correlation: as exports increase, intactness decreases. By calculating 1 - BII, you flip the scale into a metric where a higher number represents greater damage. This transformation ensures that a positive regression coefficient which is more intuitive to the reader; it also suggests that as beef export volumes go up, the associated level of environmental degradation also goes up.

Additionally, initial descriptive insights revealed that the distribution of beef exports is highly skewed [meaning a few countries export massive amounts while most export very little]. To satisfy the assumptions of linear modeling and prevent these differentials from disproportionately biasing the results, trade volumes were log transformed (Log10) [a mathematical scaling that pulls extreme outliers closer to the average to reveal underlying patterns]. Finally, the dataset was filtered to include only countries with consistent data across both sources, ensuring the analysis was based on a verified sample of 32 exporting nations.

2.3 Analytical Design

This project used a cross-sectional regression analysis based on eleven-year averages spanning 2010 to 2020. The rationale for using averages rather than raw annual data is that biodiversity loss is a cumulative, a longer term process, whereas annual trade volumes can fluctuate significantly due to short term economic shocks or shifts in things like trade deals, etc. Utilising country level averages smooths these volatile fluctuations to capture the more structural, longer term relationship between a nation’s typical export profile and ecological state. Also, by bridging historical records with projected data, the study captures a more contemporary snapshot of trade-biodiversity nexus, providing a larger

sample size that increases the statistical power [the probability that a study will detect an effect or relationship if one truly exists] of the findings.

2.4 Model Specification

The relationship was estimated using an Ordinary Least Squares (OLS) regression model [a statistical method for modeling the relationship between a dependent variable and an independent variable]. The model is specified as follows:

Log10(Export Volume) = Constant + Coefficient * (bii_loss) + Error

In this model, the log transformed volume of beef exports to the UK serves as the dependent variable [the outcome being tested/measured], while bii_loss serves as the primary independent variable [the factor supposed to influence the outcome/change]. To ensure the findings were not influenced by heteroscedasticity [a situation where the variance of the residuals is not constant, which can lead to false claims of significance], the model was estimated using HC3 robust standard errors. This choice ensures that the resulting p values [the probability that the observed relationship happened by pure chance] are conservative and reliable, providing a stronger integrity test of the respective research question.

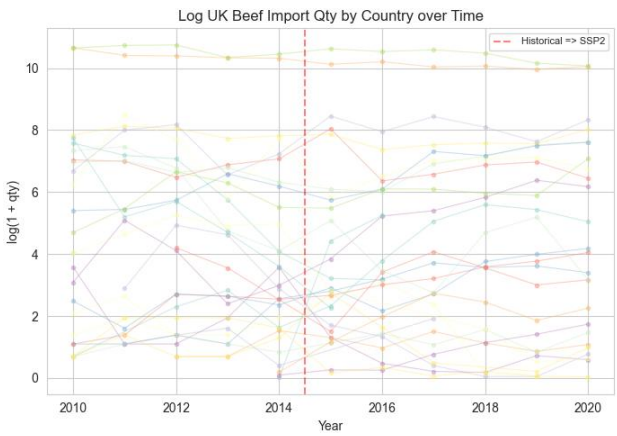
3. Results

3.1 Descriptive Insights

Intial exploration of the dataset (sample size N = 32) suggests substantial variation in both trade volumes and ecological conditions across the sample. Notably, trade data has pronounced skewness [a lack of symmetry where a small number of observations take disproportionately large values]. A limited number of dominant exporters account for the majority of beef imports into the UK, while most countries export relatively small quantities. This uneven distribution reflects the highly concentrated structure of global agricultural trade, where a few countries dominate supply chains due to comparative advantages in land availability, production capacity, and established export infrastructure. The BII also varies considerably across countries. Some exporting nations maintain relatively high levels of biodiversity intactness, while others exhibit substantial ecological degradation. The derived bii_loss variable highlights that biodiversity loss is not uniformly distributed across exporters, but instead reflects differing national contexts, including land-use intensity, agricultural expansion, and environmental governance. Taken together, these descriptive patterns suggest that the dataset captures both significant economic

inequality in trade participation and heterogeneity in ecological conditions, suggesting a suitable basis for examining whether these dimensions are systematically related.

To visualize this distribution and justify the use of a log transformation. This histogram below shows that log transformation reduces skewness and prevents extreme exporters from dominating the analysis, revealing underlying patterns in trade volumes.



3.2 Main Regression Results

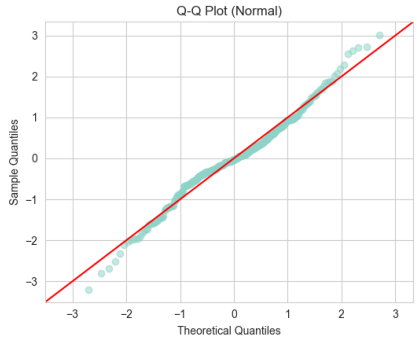
To evaluate the research question, the relationship between biodiversity loss and beef export volumes was estimated using an Ordinary Least Squares (OLS) model with HC3 robust standard errors. The results are summarised as follows:

- Correlation and Direction
  The analysis identifies a positive relationship between biodiversity loss and UK beef export volumes. The estimated coefficient for bii_loss is 8.2046 (~8.205/+8.205), indicating that a one-unit increase in biodiversity loss is associated with an approximate 8.2 increase in the log of UK beef import quantity. Given that the dependent variable is log-transformed, this relationship can be interpreted as a proportional increase in export volumes associated with increases in biodiversity loss. In practical terms, countries with higher biodiversity loss tend to export substantially larger volumes of beef to the UK.
- Statistical Significance
  The model produces a p-value of 0.0479, which falls below the conventional threshold of 0.05. This suggests that the observed relationship is unlikely to have arisen by chance under the null hypothesis of no association. In practical terms, this gives evidence/statistical support for the existence of a relationship between

biodiversity loss and export volumes within the sample, although it does not establish causality. However, because the p-value is very close to 0.05, the result should be interpreted as statistically fragile, suggesting limited robustness and sensitivity to model specification.

- Model Fit (R-squared)
  The R-squared value of 0.1408 suggests that approximately 14.1 %of the variation in beef export volumes to the UK can be explained by differences in biodiversity loss across countries. While this confirms that biodiversity loss has some explanatory power, it also implies that most of the variation in trade volumes is driven by other factors not captured in the model, such as trade policy, geographic proximity, and market dynamics.

Overall, the OLS results provide evidence of a positive but modest and fragile association between biodiversity loss and Uk beef import volumes.

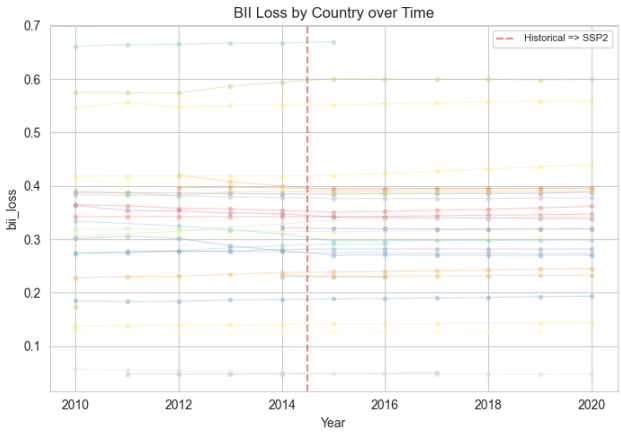


3.3 Key Patterns

First, the combination of statistical significance but a relatively low R-squared value suggests that the relationship is present but limited in strength. This suggests that biodiversity loss is one contributing factor among many. Second, the positive coefficient aligns with the expectation that higher export intensity may be associated with more land-intensive production systems, that the beef industry is, which in turn place greater

pressure on ecosystems. This pattern is particularly visible among the largest exporters in the dataset, which tend to coincide with countries exhibiting higher levels of biodiversity loss. However, the presence of notable exceptions complicates interpretation of environmental pressures at the production level. Some countries maintain relatively high biodiversity intactness while still exporting significant quantities of beef, while others exhibit biodiversity loss without corresponding export intensity to the UK. Finally, these deviations indicate that the relationship is not uniform and may be mediated by additional factors such as domestic consumption, differences in production practices, etc.

Overall, the results establish a statistically significant but modest association between biodiversity loss and UK beef export volumes. This reenforces that the observed relationship reflects a broad structural pattern rather than a universal rule across all countries.



4. Discussion

4.1 Interpreting the relationship(s)

The regression results provide mixed evidence on the relationship between biodiversity loss and Uk beef import volumes, depending on the model specification.

The OLS results indicate a positive and statistically significant relationship, with a coefficient ~8.2($p = 0.048$). This suggests that countries with higher biodiversity loss tend to export larger volumes of beef to the UK. However, this estimate primarily reflects cross-country differences, rather than changes within countries over time.

In contrast, the Fixed Effects model , which controls for all time-invariant country characteristics, produces a negative coefficient (-29.8) that is only weakly

significant(p~0.077). This implies that within a given country over time, increases in biodiversity loss are associated with reductions (or no clear increase) in export volumes.

The difference between these results is important. The positive OLS relationship appears to be driven by structural differences between countries, rather than a causal relationship within countries.

The Hausman test(p=0.042) rejects the Random Effects model in favor of Fixed Effects (FE), indicating that country-specific characteristics are likely correlated with biodiversity loss. As a result, the FE model provides the more reliable estimate.

Taken together, the evidence suggests that the observed positive relationship is likely spurious or overstated and reflects cross-country composition effects rather than true within-country causal mechanisms. This highlights the importance of accounting for unobserved heterogeneity when analyzing trade-environment relationships, as failing to do so can lead to misleading conclusions.

4.2 Explaining the Magnitude and Model Fit (R squared)

While the coefficient is statistically significant (p = 0.0479), the R-squared value of 0.1408 indicates that biodiversity loss explains only 14.1 percent of the variation in export volumes. This relatively low explanatory power is not a weakness of the model itself but could reflect the multidimensional nature of trade and its relationship to environment.

Export volumes in the FAOSTAT dataset are shaped by factors that are not included in the model, such as comparative advantage, historical trade relationships, and geographic proximity to the UK market. At the same time, the bii_loss variable is derived from national-level biodiversity data, which aggregates ecological conditions across entire countries rather than isolating the specific regions where beef production occurs. This spatial aggregation reduces the precision with which biodiversity loss can be linked to export activity, thereby limiting the proportion of variance the model can explain.

The use of cross-sectional averages over the 2010 to 2020 period further contributes to this outcome. While averaging reduces short-term volatility in trade data, it also compresses variation, particularly for countries whose export volumes fluctuate significantly over time. This smoothing effect makes it more difficult for the model to capture dynamic relationships, contributing to a lower R-squared value. The result is a model that detects a statistically meaningful trend but does not capture the full reality of the underlying system.

This interpretation is reenforced by the panel regression results. The FE model produces very low within R-squared values (~0.03-0.04). indicating that changes in biodiversity loss within countries over time explain very little of the variation in export volumes. This is consistent with the earlier finding that most variation in biodiversity loss occurs between countries rather than within.

As a result, the explanatory power observed in OLS model is likely driven by cross-country differences, rather than meaningful within-country dynamics over time.

4.3 Interpreting patterns in the data

The descriptive finding that export volumes are highly skewed plays a central role in shaping the regression results. The log transformation applied in the data preparation stage reduces the influence of extreme values, but does not eliminate the dominance of a small number of exporters. Instead, it allows these countries to remain influential while preventing them from fully determining the slope of the regression line.As a result, the positive coefficient on bii_loss reflects a pattern in which the highest-exporting countries tend to cluster at higher levels of biodiversity loss, even after scaling. This clustering effect is visible in the underlying data and is reinforced by the model.

Countries with relatively low biodiversity loss but moderate export volumes suggest that it is possible to participate in trade without equivalent levels of ecological degradation, potentially due to differences in production efficiency or environmental regulation. Conversely, countries with high biodiversity loss but low export volumes suggest that ecological degradation is not solely driven by exports to the UK, but may instead reflect domestic consumption or trade with other markets.

4.4 Strengths

A strength of this analysis lies in the direct linkage between biodiversity metrics and bilateral trade data at the country level. By constructing the bii_loss variable and aligning it with FAOSTAT export data through ISO3 standardisation, the project creates a coherent dataset that allows for empirical testing of the question. Also, without the log-controlled values, the regression would be dominated by a small number of extreme observations, obscuring any underlying relationship. Additionally, the inclusion of SSP2-based projections extends the dataset beyond the limited 2010 to 2014 overlap period, increasing the number of observations used in the averaging process. This enhances statistical power, making it more likely that a true relationship (if present) is detected by the model.

4.5 Limitations

Despite these strengths, the results must be interpreted with caution due to several limitations. First, the model does not establish causality. The positive coefficient on bii_loss indicates association, but does not demonstrate that exports to the UK directly cause biodiversity loss. Both variables may be jointly influenced by a third confounding factor such as agricultural expansion or land availability. Secondly, the bii_loss variable is derived from country-level averages of biodiversity intactness, which encourages scale mismatch. Export data reflect specific production activities, whereas biodiversity loss is measured across entire national territories. This reduces the precision of the relationship and may weaken the observed association. Thirdly, the use of SSP2 projections introduces model uncertainty. While this approach allows the dataset to extend to 2020, the projected values depend on assumptions about future socio-economic conditions that may not fully align with observed developments.Fourth, the absence of control variables creates the risk of omitted variable bias (a distortion in estimated relationships caused by excluding relevant explanatory factors). Variables such as GDP, land use intensity, and agricultural policy are likely to influence both biodiversity loss and export capacity, and their omission limits the interpretability of the coefficient.

5. Conclusion

This project set out to test the hypothesis that higher volumes of beef exports to the UK are associated with greater biodiversity loss in exporting countries. Using a cross-sectional regression model based on country-level averages from 2010 to 2020, the analysis finds a positive and statistically significant relationship between biodiversity loss (bii_loss) and export volumes. The estimated coefficient of 8.2046, alongside a p-value of 0.0479, provides sufficient statistical evidence to reject the null hypothesis of no association at the 5 percent significance level. However, while the hypothesis is supported in more statistical terms, the strength of the relationship is limited. The R-squared value of 0.1408 indicates that biodiversity loss explains only a relatively small proportion of the variation in export volumes across the sample of 32 countries. This suggests that although an association exists, it is not strong enough to suggest that biodiversity loss is a primary determinant of trade intensity. Instead the results point towards a partial relationship embedded within a much broader set factors at play. The findings suggest that countries with higher export volumes to the UK are more likely to be associated with land-intensive production systems, which are in turn linked to greater ecological degradation. This aligns with the structural characteristics of the beef sector, where production often requires large-scale land conversion. At the same time, the presence of variability and exceptions within the

dataset highlights that this relationship is not uniform and is likely mediated by factors such as policy, environmental governance, and more. Crucially, the analysis can be said to not establish causality. The observed relationship reflects an association between two aggregated national-level variables, rather than direct evidence that UK demand drives biodiversity loss in exporting countries. The results therefore support the hypothesis in a limited sense, confirming the existence of a relationship, but not its mechanism, or magnitude beyond the observed statistical association.

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